10.9 Symmetry Factors

In quantum field theory, symmetry factors correct for overcounting in perturbative expansions. This note supplements the discussion in lectures: first the definition and three standard examples, then worked examples in φ^3 theory.

Origin and Definition

Symmetry factors arise because several *Wick contractions*, or permutations of vertices and lines, can produce the same term. The redundancy is measured by the **automorphism group** of the diagram: all relabelings of vertices and of the line ends attached to them (permutations, rotations, reflections) that leave the diagram unchanged. The **symmetry factor** is the order of this group, $S = |\mathrm{Aut}(G)|$, and each diagram is weighted by $1/S$.

Three Examples

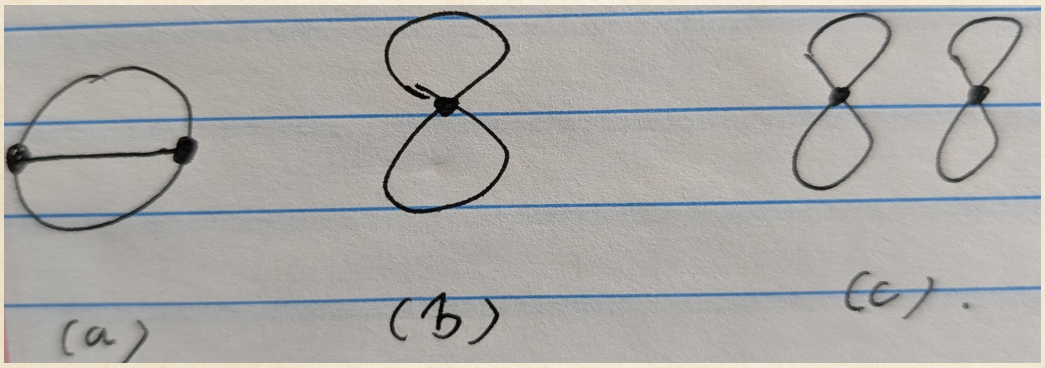


Diagram (a) is in φ^3 theory; (b) and (c) are vacuum diagrams of φ^4 theory, where each vertex has four lines.

- **The "sunset" diagram in φ^3 theory** (a): two vertices connected by three lines.
  - Symmetries: vertex swap ($\mathbb{Z}_2$) and permutations of the three lines (S_3).
  - $|\mathrm{Aut}(G)| = 2 \times 6 = 12 \Rightarrow S = 12$.
- **The vacuum "figure-eight" in φ^4 theory** (b): one vertex with two loops.
  - Symmetries: swapping the loops ($\mathbb{Z}_2$) and flipping each loop ($\mathbb{Z}_2 \times \mathbb{Z}_2$).
  - $|\mathrm{Aut}(G)| = 2 \times 2 \times 2 = 8 \Rightarrow S = 8$.
- **Two identical disconnected pieces** (c): two copies of the figure-eight.

- Symmetries: swapping the two pieces ($\mathbb{Z}_2$) and the symmetries of each piece.
- $|\mathrm{Aut}(G)| = 2 \times (8 \times 8) = 128 \Rightarrow S = 128$.

Worked Examples in φ^3 Theory

Recall how we "got rid of" the factor

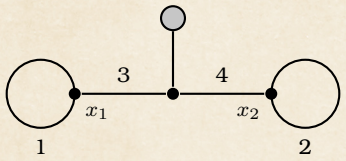
$$\frac{1}{V!}\left(\frac{1}{3!}\right)^V \frac{1}{P!}\left(\frac{1}{2!}\right)^P$$

in the expansion of $Z(J)$ (Srednicki §9). We treated the V vertices and the P propagators as identical and exchangeable, which gives $V!\,P!$ rearrangements. The three derivatives $\left(\frac{1}{i}\frac{\delta}{\delta J}\right)^3$ of each vertex, acting on the factors $\frac{1}{2}J\Delta J$, then supply another $(3!)^V(2!)^P$ possibilities, cancelling the factor exactly. But this overcounts whenever some of those rearrangements give back the same term. The overcounting is the symmetry factor S, and each diagram is weighted by $1/S$. Shaded circles below are sources.

Example A: the sunset

The sunset, diagram (a) above, has $S = 2 \times 3! = 12$: its three lines can be permuted ($3!$) and its two vertices exchanged (2). Srednicki reaches the same 12 in §9 by counting which rearrangements duplicate one another (the second diagram of his Fig. 9.1, below (9.11)), and "The count for $V = 2$, $P = 3$" below counts its 288 terms.

Example B: two tadpoles on a line, one source



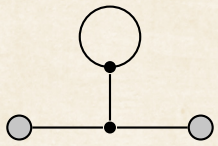
(1) The loops 1 and 2 are $\Delta(x_1 - x_1)$ and $\Delta(x_2 - x_2)$. Flipping the ends of a loop gives back the same term; there is no $\Delta(x - y) \to \Delta(y - x)$ to tell the two orders apart. So the weight is overcounted by 2×2.

(2) Exchanging propagators 3 and 4 together with the vertices x_1 and x_2 gives the same process. It only changes how the functional derivatives are arranged and produces no new term, so it overcounts by another 2.

(3) The symmetry factor is $S = 2 \times 2 \times 2 = 8$.

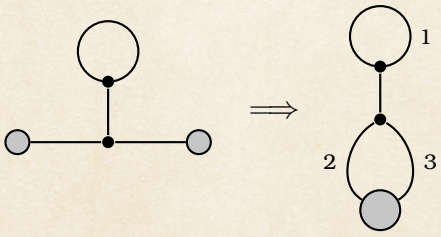
(4) By symmetry: a left–right reflection, and an up–down reflection of each of the two loops, give $S = 2 \times 2 \times 2 = 8$ again.

Example C: a tadpole on a line between two sources



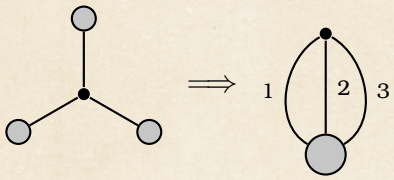
(1) The loop gives an overcounting of 2.
(2) The left–right reflection gives another 2.
(3) $S = 2 \times 2 = 4$.

For several sources there is a useful *deformation method*. Since the sources are identical, deform the diagram so that all of them meet in one shaded blob. The blob is not a vertex: any number of lines may end on it. Then read off the symmetries of the deformed diagram. For Example C:



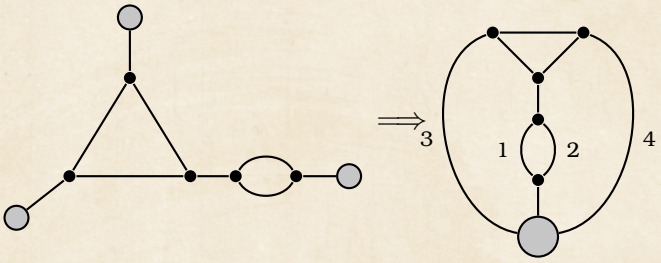
Propagator 1 gives a factor of 2, and propagators 2 and 3 can be exchanged ($2!$). So $S = 2 \times 2 = 4$.

Example D: three sources on one vertex



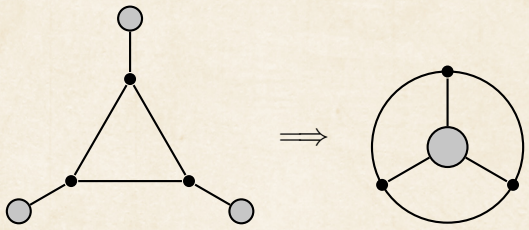
The three propagators can be permuted in every way, so $S = 3! = 6$.

Example E: a triangle with a bubble on one leg



After the deformation one can exchange $1 \leftrightarrow 2$ (the two lines of the bubble) and $3 \leftrightarrow 4$ (a reflection of the triangle that fixes the dressed corner). So $S = 2 \times 2 = 4$.

Example F: a triangle with three bare legs



The deformed diagram has a cyclic symmetry of order 3 (rotations by $120°$ and $240°$) and a reflection. The three reflections, one through each corner, are not independent: each is a rotation times one reflection. So $S = 3 \times 2 = 6$.

Why $1/S$: Terms, Diagrams and Relabelings

At the sunset (the theta of LecNote8 p. 4) the margin asks: "we only have one possibility! How can I count it $2! \times 3!$ times?" And where did the $1/3!$ of each vertex and the $1/2$ of each line go? With $Z_g = 1$ and $\mathcal{L}_{\rm ct}$ dropped, LecNote8 p. 3 expands

$$Z_1(J) \propto \sum_{V=0}^{\infty} \frac{1}{V!}\Big[\frac{ig}{3!}\int d^4x \Big(\frac{1}{i}\frac{\delta}{\delta J(x)}\Big)^3\Big]^V \times \sum_{P=0}^{\infty} \frac{1}{P!}\Big[\frac{i}{2!}\int d^4y\, d^4z\, J(y)\Delta(y-z)J(z)\Big]^P, \quad (10.9.1)$$

with Srednicki's Δ, $(-\partial^2 + m^2)\Delta = \delta^4$. Each of the $3V$ derivatives hits a different one of the $2P$ factors J; the $E = 2P - 3V$ left over are sources. A **term** is one such assignment, and there are $(2P)!/E!$ of them. A **diagram** D is the class of all terms that draw the same picture.

Renaming the V vertex variables, permuting the three derivatives at each vertex, permuting the P propagator factors and swapping the two ends of each (Δ is even) leave Eq. (10.9.1) unchanged. These relabelings form a group R of order

$$|R| = V!\,(3!)^V\,P!\,(2!)^P, \quad (10.9.2)$$

exactly the inverse of the prefactor: that is where the $1/3!$ and the $1/2$ go. A relabeled term has the same value (it is the same integral with renamed variables), and two terms draw the same picture exactly when a relabeling carries one into the other. So the terms of D form one orbit of R, and by the orbit–stabilizer theorem the orbit has $|R|/|\mathrm{Stab}_D|$ terms, where Stab_D, the relabelings that map a term to itself, is $\mathrm{Aut}(D)$ of "Origin and Definition". Times the prefactor $1/|R|$, the diagram contributes its value divided by $|\mathrm{Aut}(D)| = S$. So $1/S$ is not put in by hand: it is what is left of the Taylor coefficients once identical terms are added up.

The count for $V=2$**,** $P=3$**.** The six derivatives hit the six J's in $6!=720$ ways. For the sunset, vertex 1 sends its legs to three different lines ($3!$) and picks an end of each (2^3), and vertex 2 fills the other ends ($3!$): $3!\cdot 2^3\cdot 3!=288$ terms. For the dumbbell, one leg at each vertex goes to the bar ($3\cdot 3$), the lines take the roles bar, loop at vertex 1, loop at vertex 2 ($3!$), and the ends are filled (2^3): 432 terms, and $288+432=720$. With $|R|=2!\cdot 6^2\cdot 3!\cdot 2^3=3456$, the coefficients are $288/3456=1/12$ and $432/3456=1/8$. The margin's $2!\times 3!$: of the 3456 relabelings, 12 map a given sunset term to itself (a permutation of the lines, with or without a swap of the vertices, each with the matching legs and ends); the other 3444 carry it to the other 287 sunset terms, 12 to each.

A Completeness Test: the Sum Rule

"How do I know that my list of diagrams (LecNote8 pp. 4–5) is complete, and every S right?" Every term belongs to exactly one diagram, so $\sum_D |R|/S_D=(2P)!/E!$. With $2P=3V+E$ and $(2P)!/(P!\,2^P)=(2P-1)!!$,

$$\sum_D \frac{1}{S_D}=\frac{(3V+E-1)!!}{V!\,6^V\,E!}, \tag{10.9.3}$$

summed over all diagrams with V vertices and E sources, connected or not. The numerator is the number of Wick pairings of the $3V$ legs and E sources. A forgotten diagram or a wrong S breaks the equality. The table below checks the cases of LecNote8: in every row the last column equals the right side, e.g. $15/(2\cdot 36)=5/24$ for $(0,2)$ and $10395/(24\cdot 6^4)=385/1152$ for $(0,4)$.

(E,V)	connected S	disconnected S	$\sum_D 1/S_D$
$(0,2)$	12 (sunset), 8 (dumbbell)	—	$5/24$
$(1,1)$	2 (tadpole)	—	$1/2$
$(3,1)$	6 (Example D)	4	$5/12$
$(2,2)$	4 (Example C), 4	8, 16, 24	$35/48$
$(1,3)$	8 (Example B), 4, 4	16, 24	$35/48$
$(0,4)$	24, 48, 16, 16, 8	128, 96, 288	$385/1152$

The sum rule for the diagrams of LecNote8.

The disconnected S are products over the pieces, with an extra $2!$ for two identical pieces (next subsection). The other unnamed connected ones: the bubble on a line at $(2,2)$; at $(1,3)$, a source on a triangle with one side doubled, and a source line through a bubble into a tadpole; at $(0,4)$, the tetrahedron (24), three tadpoles on one vertex (48), a tadpole–bubble–tadpole chain and a ring of two bubbles (16 each), and a tadpole on a triangle with one side doubled (8).

Disconnected Diagrams and $Z = e^{iW}$

LecNote8 pp. 7–8: why do n identical pieces carry $1/n!$, and why should the *logarithm* of Z be connected? Let D consist of n_I copies of each connected diagram C_I. An automorphism maps every connected piece onto an identical one, so it permutes identical copies and acts on each:

$$|\mathrm{Aut}(D)| = \prod_I n_I!\, S_I^{n_I}. \tag{10.9.4}$$

Diagram (c) above is one case ($2 \cdot 8 \cdot 8$); in φ^3 theory, two sunsets give $2! \cdot 12^2 = 288$, two dumbbells $2! \cdot 8^2 = 128$, a sunset and a dumbbell $12 \cdot 8 = 96$. With C_I also standing for the value of the connected diagram, $1/S_I$ included, D contributes $\prod_I C_I^{n_I}/n_I!$: the extra factor S_D of LecNote8 p. 7 is $\prod_I n_I!$, not its inverse, and it divides. Summing over every n_I independently,

$$Z_1(J) \propto \sum_{\{n_I\}} \prod_I \frac{C_I^{n_I}}{n_I!} = \exp\Big(\sum_I C_I\Big), \tag{10.9.5}$$

and once the vacuum diagrams are divided out (p. 8; what they are and why this is allowed is in §10.10), $Z = e^{iW}$ with iW the connected diagrams with sources (p. 9).

This is no accident of the algebra. In probability the logarithm of a generating function generates the cumulants, and a Gaussian of zero mean has only one, its variance; correspondingly $\ln Z_0 = \frac{i}{2}\int J\Delta J$ is a single connected diagram. The physical reason is **cluster factorization**: for sources J_1, J_2 far apart in space, the two experiments cannot influence each other, so $Z(J_1 + J_2) \to Z(J_1)Z(J_2)$ and W must be additive. Products of separated pieces cannot appear in W; its terms that involve both J_1 and J_2 must die off as the sources are pulled apart. A connected diagram does, because it contains propagators across the gap, and these fall off like e^{-mr} at spacelike distance r. In the free theory the only cross term is $i\int J_1\Delta J_2$; for point sources j_1, j_2 held on for a long time T it is $ij_1j_2T\, e^{-mr}/4\pi r$, since $\int dt\, \Delta(t,\mathbf{r}) = e^{-mr}/4\pi r$, the Yukawa potential.

Labelled External Lines

LecNote9 p. 3 finds the $4!$ ways to apply $\delta_1\delta_2\delta_3\delta_4$ to fall into "3 groups . . . each with 8 identical diagrams" and calls the cancellation of S "a general result for TREE-level diagrams"; rule 8 on p. 7 keeps a factor for loops. Why, and when does one survive?

$\langle 0|T\varphi(x_1)\cdots\varphi(x_E)|0\rangle$ comes from $\frac{\delta}{i\delta J(x_1)}\cdots\frac{\delta}{i\delta J(x_E)}$ acting on Z. On a diagram with E sources and weight $1/S$, the derivatives remove the sources in $E!$ ways, each putting the labels $x_1,\ldots,x_E$ on the ends. Two assignments give the same labelled diagram exactly when an automorphism relates

them, and those that fix an assignment fix every source; call their number $S_{\rm lab}$. By orbit–stabilizer, each labelled diagram arises $S/S_{\rm lab}$ times, there are $E!\,S_{\rm lab}/S$ of them, and each has weight

$$\frac{1}{S}\cdot\frac{S}{S_{\rm lab}}=\frac{1}{S_{\rm lab}}. \tag{10.9.6}$$

These orbits are the "topologically inequivalent" diagrams of LecNote9's rule 2 (p. 6).

Trees have $S_{\rm lab}=1$. Removing a vertex of a tree leaves three branches, each ending at an external point, so every vertex lies on the path between two external points. Paths in a tree are unique, so an automorphism that fixes the external points fixes every vertex and line, and (with no loops or multiple lines) every leg. The H of LecNote9 p. 3 has $S=8$, the swaps in the page's parenthesis, so $4!=3\times 8$: three labelled diagrams, the s, t and u channels, each with weight 1.

Loops can keep a symmetry with the ends fixed. The bubble on a line goes from $S=4$ (swap the bubble's lines; reflect end to end) to $S_{\rm lab}=2$, the $\frac{1}{2}$ of the one-loop self-energy (LecNote12 p. 2); Example C also goes from 4 to 2. In $2\to 2$ at one loop the box goes from 8 to 1 (three labelled boxes), the triangle at one vertex from 4 to 1 (six), and the bubble on the internal line from 16 to 2 (three); §10.11 lists all the one-loop $2\to 2$ diagrams. That is rule 8: a factor survives only from a loop symmetry that leaves every external point in place.

Zero Dimensions: the Vacuum Sums as a Series

Does the sum over diagrams converge? The simplest place to ask is zero dimensions, where the field is a single number, $\Delta=1$ and there are no integrals, so a diagram is worth $g^V J^E/S_D$. Expanding

$$Z(g,J)=\frac{1}{\sqrt{2\pi}}\int d\varphi\, e^{-\varphi^2/2+g\varphi^3/6+J\varphi} \tag{10.9.7}$$

with $\langle\varphi^n\rangle=(n-1)!!$ gives the right side of Eq. (10.9.3) as the coefficient of $g^V J^E$, so $Z=\sum_D g^V J^E/S_D$, a formal series (the integral diverges for every real $g\neq 0$). The vacuum sums of the table are the coefficients $c_k=(6k-1)!!/\big[36^k(2k)!\big]$ of g^{2k} in

$$Z(g,0)=1+\tfrac{5}{24}g^2+\tfrac{385}{1152}g^4+\cdots,\qquad \ln Z(g,0)=\tfrac{5}{24}g^2+\tfrac{5}{16}g^4+\cdots,$$

and the logarithm keeps only the connected ones: the disconnected $\frac{1}{288}+\frac{1}{128}+\frac{1}{96}=\frac{1}{2}(\frac{5}{24})^2$ is the second-order term of the exponential. The pairings outrun the prefactor, $c_{k+1}/c_k\to 3k/2$: factorial growth and zero radius of convergence, as for φ^4 in §10.17, but with all $c_k>0$ where φ^4's alternate.

SUMMARY

The contribution of a diagram to the amplitude is

$$\text{Contribution} = \frac{1}{S} \times (\text{couplings}) \times (\text{integrals}).$$

Symmetry factors ensure that each distinct term is counted exactly once. Errors in S propagate to observables such as cross sections, so getting them right matters. I directly write down the symmetry factors of many diagrams in my lecture notes and provide additional notes explaining different ways to get them. There is ultimately one method: carry out the functional-derivative expansion of the path integral. There, you won't need to introduce symmetry factors, but rather count how many ways there are to get the identical expression.